

Example 1. Solve the system

$$7x + 3y \equiv 10 \pmod{16}$$

$$2x + 5y \equiv 9 \pmod{16}.$$

Solution. Recall, multiplication by an integer modulo 16 is reversible provided

Let us solve for one variable in terms of the other in the first congruence.

We substitute into the second congruence.

Example 2. Solve the system

$$7x + 3y \equiv 10 \pmod{16}$$

$$x + 3y \equiv 9 \pmod{16}$$

Solution. Just as with the last system, solving for y in the first congruence yields

We substitute into the second congruence.

Recall that $ax \equiv b \pmod{n}$ has a solution if and only if:

A. **Fermat’s Little Theorem.** Fermat was a 17th century monster of mathematics. He pursued the subject as a pastime, in correspondence with mathematicians. To number theorist Frénicle he revealed his “Little Theorem”.

This is contrasted with this “Last” or “Great Theorem”, which was not completely proven at the end of the 20th century.

Fermat’s Little Theorem. Let p be prime and:

$$\text{Then: } a^{p-1} \equiv 1 \pmod{p}$$

Proof. First we establish a lemma.

Lemma. The integers:

are congruent modulo p , perhaps after reordering, to:

Proof. Evidently the number of integers in the list are the same.

So we need only show that every integer in the second list is part of the first list.

To this end, let integer k between 1 and $p - 1$ be given.

Since $a \not\equiv 0 \pmod{p}$ it follows that:

□

Now we consider the product modulo p

$$a \cdot 2a \cdot 3a \cdots (p-1)a \equiv$$

On the other hand, using the lemma, this product modulo p is also

$$a \cdot 2a \cdot 3a \cdots (p-1)a \equiv$$

Therefore

Next, note that, because p is prime:

Hence $(p-1)!$ has an inverse, which, when multiplied to both sides yields

□

If we multiply both sides of Fermat’s little congruence by a we obtain

Corollary. Let p be prime and a be an integer. Then:

$$a^p \equiv a \pmod{p}$$

Example 3. Find the remainder when 11^{90} is divided by 23.

Solution. Because 23 is prime and $11 \not\equiv 0 \pmod{23}$ we have by Fermat’s Little Theorem

Now we compute

$$11^{90} \equiv$$

Example 4. Let a be an integer.

Show that the the last digit of a^{99} is the same as the last digit of a .

Solution. This is equivalent to showing

Because $10 = 2 \cdot 5$ and $2 \nmid 5$ this is equivalent to showing

The first congruence is easier to confirm, as either

Let’s address the second congruence. First, consider the case:

Next consider the case:

By Fermat’s Theorem

On the other hand, we investigate the exponent modulo 4.

$$99 \equiv$$

$$\text{And thus: } 99 \equiv$$

$$\text{So: } a^{99} \equiv$$

□

Example 5. Let p be an odd prime. Prove that:

$$\sum_{k=1}^{p-1} k^p \equiv 1^p + 2^p + 3^p + \cdots + (p-1)^p \equiv 0 \pmod{p}$$

Solution. By the corollary of Fermat’s Theorem

and thus

$$1^p + 2^p + 3^p + \cdots + (p-1)^p \equiv$$

Next we use the sum formula

$$1 + 2 + 3 + \cdots + n =$$

To continue

$$\cdots \equiv$$

But we assumed p is an **odd** prime, and thus:

$$\cdots \equiv$$

□